

Chain Rule

Section 2.5

A. The Chain Rule

$$[h(g(x))]' = h'(g(x)) \cdot g'(x)$$

Alternative Notation

$$f(x) = (h \circ g)(x) = h(g(x)) \text{ then } f'(x) = [h'(g(x))] \cdot [g'(x)]$$

$$[h(g(x))]' = \frac{d(h(g))}{dx} = \left(\frac{dh}{dg}\right) \cdot \left(\frac{dg}{dx}\right)$$

In Plain English, first, identify which function is on the “outside” and which is on the “inside.” (For the composition $(f \circ g)(x) = f(g(x))$ we say that f is on the “outside” and g is on the “inside.”) The derivative of this composition is equal to **the derivative of the outside (leave the inside alone) times the derivative of the inside.**

Examples

1. $f(x) = \sin(5x^5)$

First, let us identify which is on the “outside” and which is on the “inside”: Here $\sin(\dots)$ is the “outside” (i.e., “sin of something”) and $5x^5$ is the “inside.”

Derivative of the “outside” is $\cos(\dots)$, and if we leave the inside alone, this will be $\cos(5x^5)$

Derivative of the “inside” is $25x^4$

$$f'(x) = \underbrace{(\cos(5x^5))}_{\text{Derivative of the outside (leave the inside alone)}} \times \underbrace{(25x^4)}_{\text{Derivative of the inside}}$$

Note: This style of answer should only be “cleaned up” if you are given specific instructions to do (or if you have to compare it to a list of multiple choice answers)!

Again, do not forget to label your derivative

More Examples

2. $g(x) = \sin^2 x = [\sin x]^2$

We see that $g(x)$ consists of $[\dots]^2$ as the “outside” and $\sin x$ as the “inside.”

So, the derivative of the “outside” is $2[\dots]$ and derivative of the “inside” is $\cos x$

$$\Rightarrow g'(x) = 2[\sin x] \cdot [\cos x]$$

Examples: Find and LABEL the derivatives of each of the following functions:

1. (video) $f(x) = (x^2 + 3)^2$

2. (video) $h(x) = (x^3 + 5x + 7)^9$

3. (video) $g(x) = \sqrt{5x^2 + 11x - 2}$

4. (video) $k(x) = \sin(7x^3 + 3x - 5)$

5. (video) $m(x) = \frac{5}{\sqrt[3]{x^2 + 3 \cos x + 17}}$

6. $f(x) = \tan(6x^3)$

7. $k(x) = \tan(\sin x)$

8. $l(x) = \sqrt{x^2 + 1}$

9. $f(x) = (x^2 + 2x)^8$

10. $r(s) = \frac{1}{\sqrt[3]{s^4 + s}}$

B. Combinations of Product, Quotient, and Chain Rule

In many problems, we need to use a combination of the Product, Quotient, and Chain Rule to find a derivative. Here we will work through lots of examples.

Examples: Find and LABEL the derivatives of each of the following functions. Do not clean up unless otherwise indicated.

11. $g(x) = \sin(x^2)(8x^3 - 1)$

12. $f(x) = \sin^3(x^2)$

13. $g(x) = (x - \cos^2 x)^4$

14. $r(\theta) = \frac{\cos(2\theta - 4\pi)}{\theta^2}$

15. $a(x) = \left(x^2 \sqrt{2x^3 + 1}\right)$

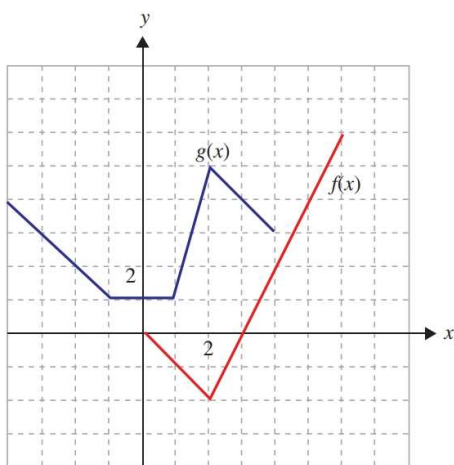
16. $f(x) = \frac{5x^3}{(x^2 + 5x)^3}$

17. If $f(x) = \cos(a^8 + x^8)$, then $f'(x) =$

18. If $f(x) = \cos(x \sin x)$, find $f'(x) =$

19. Let $F(x) = f(g(x))$, where $f(-7) = 2$, $f'(-7) = 2$, $f'(3) = 15$, $g(3) = -7$, and $g'(3) = -8$, find $F'(3)$

20. If f and g are the functions whose graphs are shown below, let $u(x) = f(g(x))$ and $v(x) = g(f(x))$.



Find $u'(3)$

and $v'(3)$

21. Let $F(x) = f(x^6)$ and $G(x) = (f(x))^6$. You also know that $a^5 = 11$, $f(a) = 2$, $f'(a) = 8$, $f'(a^6) = 15$.

Find $F'(a)$

and $G'(a)$